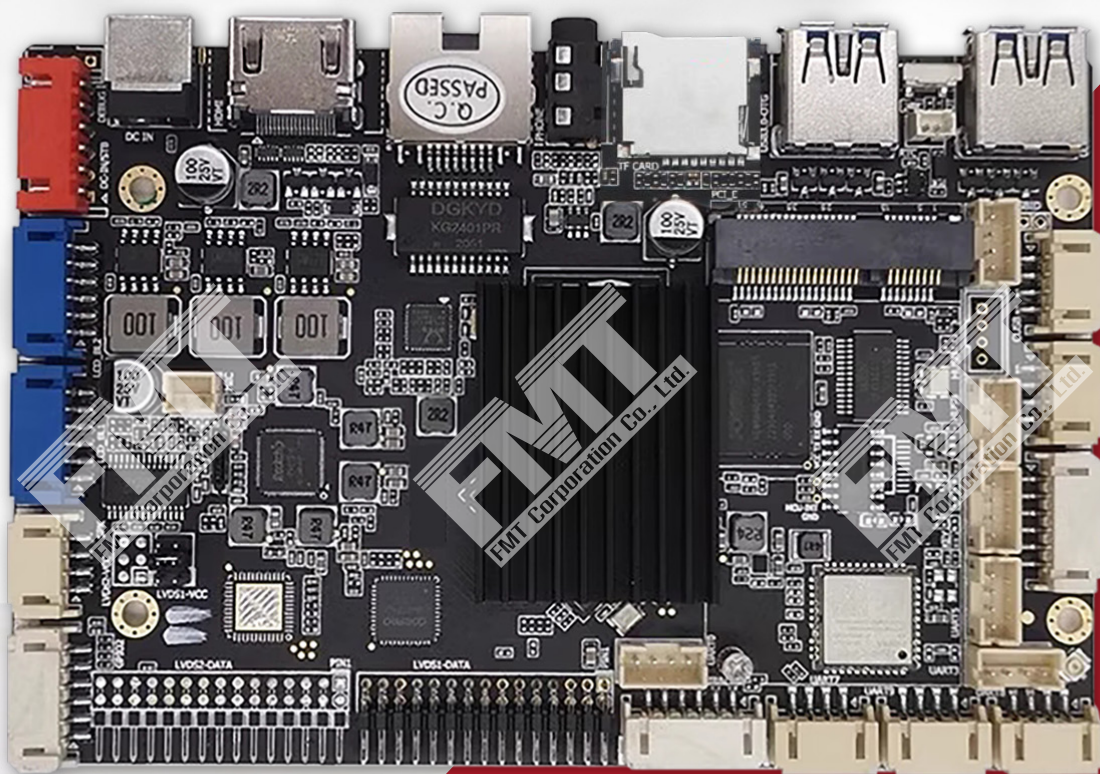




▪ Overview

FM-3568D uses Rockchip RK3568 (Cortex-A55) high-performance, low-power quad-core processor, equipped with Android 11 system, main frequency up to 2.0GHz. embedded GPU (ARM G52 2EE) supports OpenGL ES 1.1/2.0/3.2, OpenCL 2.0, Vulkan 1.1. Integrated HDMI/LVDS/EDP/MIPI display interface, Ethernet/WIFI/BT/4G (requires additional 4G module) expansion interface in one, greatly simplifying the entire machine design. Support most of the current popular video and picture format decoding, high-performance VPU support support 4K 60fps H.265/H.264/VP9 video decoding, support 1080P 60fps H.265/H.264 video encoding support 8M ISP, support HDR. can drive a variety of TFT LCD display. More stable, widely used in AI servers, face payment devices, security, medical, transportation, finance, industrial control, intelligent education, intelligent retail and other AI intelligent fields.

▪ Feature

High integration : Integrated dual LVDS or dual MIPI, EDP and HDMI display interfaces in one, which can support up to three different displays and simplify the whole machine design, and can be inserted into TF card to modify the screen parameters.

High stability : In terms of hardware and software, we add our own self-developed technology to ensure the stability of the product, which can make the final product reach 7x24 hours unattended.

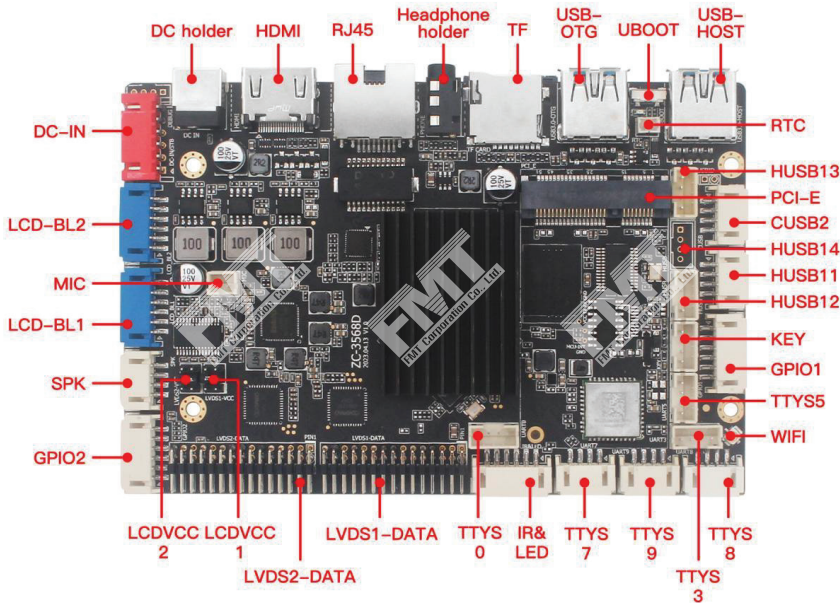
Rich expansion interfaces : 6 USB interfaces (default 2 pins can be changed to 4 pins, 2 standard USB3.0), 6 expandable serial ports (default 3 can be changed to 6), GPIO/ADC interface, which can meet the requirements of various peripherals in the market.

High definition : support up to 3840 x 2160 decoding and various LVDS/EDP interface LCD displays.

Complete functions : support dual-screen different display, horizontal and vertical screen playback, video split screen, scrolling-subtitles, timing switch, USB data import and other functions.

Convenient management : humanized playlist background management software for easy management and control of ad playback. Playback log is easy to understand the playback status

PCB-Front Layout



PCB-Back Layout

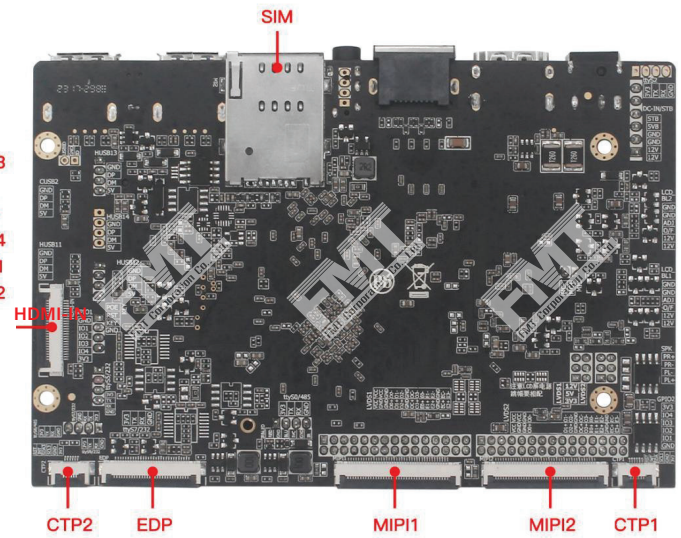
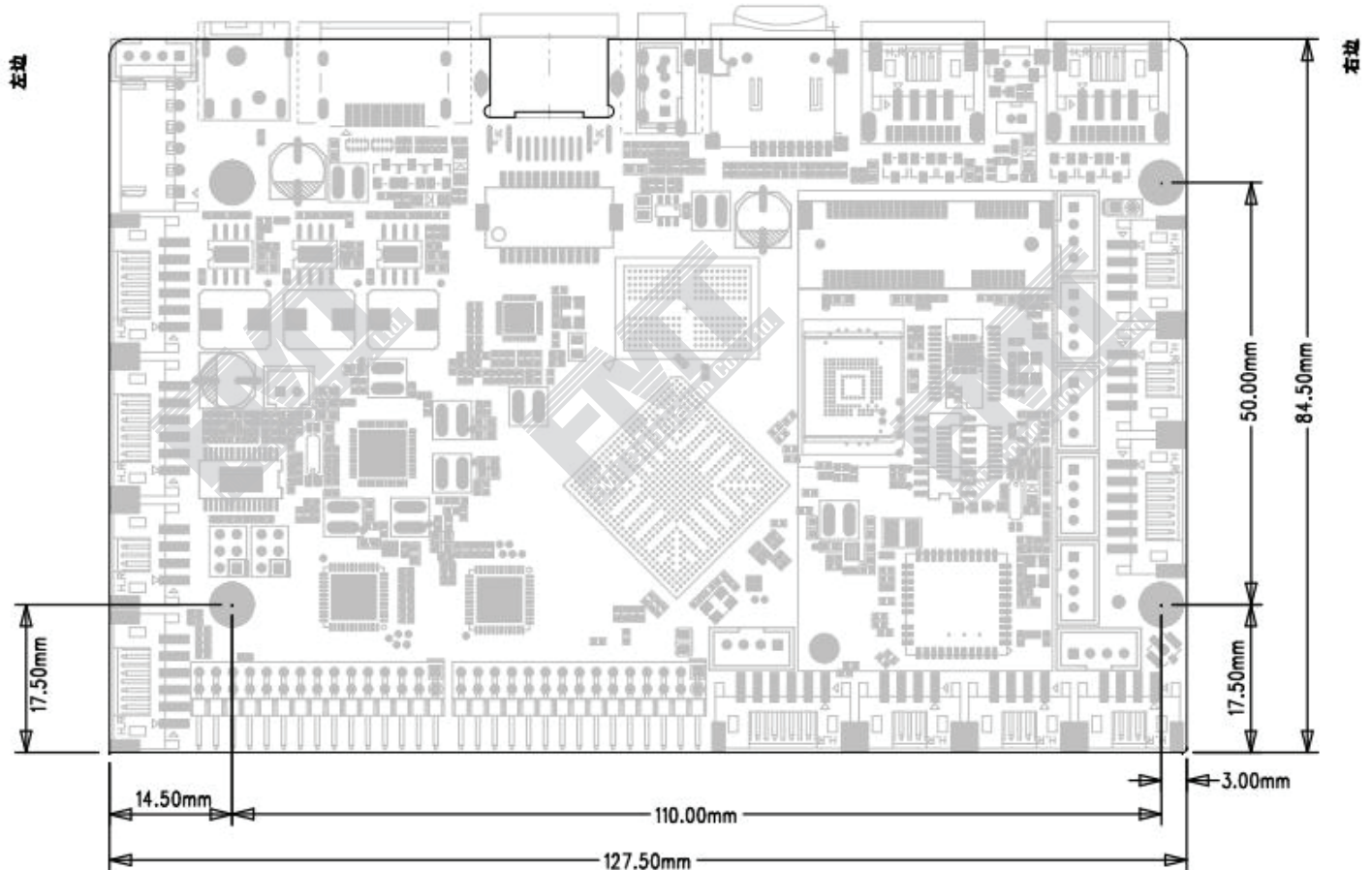


Photo statement : The above pictures are taken from a certain batch of boards (a certain combination of functions) of our company. Since the products are constantly being updated and maintained, the actual boards shipped may not be consistent with the photos. The samples obtained by the customers are based on the samples. shall prevail.

PCB-Dimension



PCBA size: 127.5mm x 84.50mm/± 0.5mm; Plate thickness 1.6mm ± 10%

Screw hole specification: ∅ 3.0mm x 4/± 10%

Motherboard Specification

CPU	Quad-core 64-bit Cortex-A55, up to 2.0 Ghz
GPU	ARM G52 2EE supports OpenGL ES 1.1/2.0/3.2, OpenCL 2.0, Vulkan 1.1
NPU	Computing power up to 1 TOPS
Memory	2G/4G/8G (optional)
Internal Memory	EMMC 16G / 32G/ 64G/128G(optional)
Operating System	Android 11
Playback Mode	Support loop, timing, interpolation and other playback modes
Video format support	wmv, avi, flv, rm, rmvb, mpeg, ts, mp4, etc.
Picture format support	BMP, JPEG, PNG, GIF
Power supply interface	1 built-in 2.54MM 6P power input connector and 1 external diameter 5.5 core 2.0 DC head connector
LVDS output	2x single/dual 30 PIN Dupont interface, can directly drive 50/60Hz multiple resolution LCDs
MIPI output	2 x 4 LIN MIPI (40P 0.5MM pitch FPC interface), can directly drive 2 MIPI LCDs
EDP output	1 x 2 LIN EDP screen (30P 0.5MM pitch FPC interface), supports up to 1920*1080 output
HDMI output	1x A-type interface, support 1080P, 2K, 4K output
TF card	Support 16G/32G/64G/128G
CTP interface	2-way 12C touch screen interface, can drive two I2C touch screen use at the same time
Remote control and indicator	Support infrared remote control key control, support the indication of the device working status light
Serial port/expansion	6-channel TTL, 8-channel GPIO
Audio and video	14-wire headphone interface (U.S. standard), 1 dual-channel speaker output interface (support for left and right channel output of 8 R / 5W amplifier)
Audio input	1 microphone interface (2 PIN pins)
RTC real time clock	Support and also supports scheduled power on and off
USB interface	2x USB 3.0, 5x built-in USB socket
System Upgrade	Support local USB upgrade, wireless upgrade, computer upgrade
Network Support	- Support 10/100M adaptive Ethernet - Built-in WiFi, Bluetooth, WIFI support hotspot sharing - Extendable 4G Internet access (need another USB 4G module)
Size	127.5 x 84.50 mm. (±0.5 mm.), plate thickness 1.6mm±10%
Operating Environment	Temperature : -20°C ~ 70°C, recommended : 5°C~35°C Working Humidity : 10% ~ 90% no condensation

PCB Specification

PCB board	FR4 6-layer board sinking gold impedance board TG150 matte black
Electronic materials	Lead free, halogen-free and environmentally friendly materials (original and genuine, meeting ROHS requirements)
Manufacturing Process	Lead free, environmentally friendly production process (in accordance with ISO9001 production quality management system requirements)

Please Note :

1. Please do not exceed a total current of 2.0A on the 6 USB ports (USB-OTG, USB3.0, USB2 is the same 5V, and the LVDS converted from USB1 and RGB is the same 5V). When connecting multiple USB settings, allocate the power reasonably. USB1 can mount devices with relatively high current.
2. Please do not exceed 1A for the total current on 3.3V
3. There is a short circuit between INT and GND at the bottom of the motherboard, which is a remote control learning function. If you have any requirements, you can contact our business representative to obtain a DXF structure diagram.
4. Cooperate with our OPS small board to facilitate interconnection and interoperability with OPS supported boards.